

miniRUEDEI Symposium 18–20 Nov 2025

Eawag, Überlandstrasse 133, 8600 Dübendorf, Switzerland

Tentative/Draft Program

Each presentation:

20 min talk

5 min questions and discussion

	Tuesday	Wednesday	Thursday
9:00–9:30		<i>Coffee & Snacks</i>	
9:30–10:40	<i>Welcome</i> Bekaert Musy	Helbing Peel Brennwald	Kipfer Tomonaga Marion
	<i>Break with Coffee & Snacks</i>		
11:10–12:00	Cateland West	Seltzer Blanc	Burghard Lefevre
	<i>Lunch Break</i>		
14:00–15:10	Giroud Lightfoot Selvadurai	Bugher / Harvey Schilling / van Rooyen Welham	Gumindoga Moser Röggla Aardema
	<i>Break with Coffee & Snacks</i>		
15:40–16:30	Coppens Nuriel	Arruda Alshareef	<i>End</i>
	<i>Group Dinner</i>		

Photosynthesis and productivity rates in the Eastern Tropical Pacific Ocean over the 2023–2025 El Niño-Southern Oscillation cycle

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The El Niño-Southern Oscillation (ENSO) is a natural climate variability associated with strong fluctuations in the ocean environment that impact biota and productivity. This provides suitable conditions for testing the response of ecosystems to rapid climate change. We conducted five expeditions in the Eastern Tropical Pacific to study the impact of this climate variability on phytoplankton photophysiology and productivity. On the S/Y Eugen Seibold the underway measurement set-up includes a LabSTAF active fluorometer (Chelsea, UK) and a miniRUEDI mass spectrometer (Gasometrix, Switzerland) to estimate surface ocean photophysiology and dissolved oxygen-to-argon ratios, respectively. These two methods allow for the estimation of the two end members of productivity rates in the ocean, photosystem II electron transport rates and net community productivity rates. Our preliminary results show that during the El Niño phase phytoplankton standing stocks and volumetric photosynthesis rates are strongly declining. The net productivity rates of the communities do not reflect these declines, but instead are spatially and seasonally decoupled, which could modulate the efficiency of organic matter export to the deep ocean.

Evaluating the effects of multiple variable on the dissolution of CO₂

Mohammed Alshareef, Darren Hillegonds, Chris Ballentine

University of Oxford

In a controlled tube experiment, we simulated the injection of CO₂ through a vertical water column to investigate how bubble size and injection flux influence CO₂ dissolution dynamics. Continuous gas composition data were collected from the headspace using a miniRUEDI, allowing real-time monitoring of CO₂ concentrations. Analysis of the time-series data reveals distinct signatures associated with each variable, providing insights into the relationship between bubble dynamics, gas dissolution efficiency, and flux behavior.

NOBLE GAS CONCENTRATIONS IN A BRAZILIAN SEMIARID AQUIFER

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The limited recharge rate in confined and deep aquifers in semiarid regions constitutes an important setting for paleoclimatic studies. The Açu Aquifer is the main aquifer system responsible for the water supply in the Potiguar Sedimentary Basin, located in the equatorial margin of South America. Preliminarily, three hydrochemical groups were identified in the study area, enabling the distinction between recharge zones and confined zones at greater depths. ^{14}C was used as a qualitative age tracer, and the results showed convergence with this configuration, presenting more enriched pMC values in regions close to the recharge areas. At a preliminary stage, it was possible to identify long flow paths and a restricted recharge zone in the southernmost part of the study area. Initial interpretations of noble gas (NG) concentrations indicated the distinction of two groups: the first characterized by high excess air and low NG temperatures; and the second by low excess air and high NG temperatures. This variation is consistent with patterns generally observed in groundwater from semiarid regions, where records indicate wetter periods with higher recharge rates. The main challenge of this study, however, lies in interpreting paleotemperatures in regions influenced by mantle contributions, as indicated by the NG concentrations and the $^3\text{He}/^4\text{He}$ ratio. In addition, the horst and graben structures formed during the basin's opening, as well as fault systems reactivated after sediment deposition, may have contributed to the migration of formation waters, oil, and gas. Despite these complexities, a conceptual model of hydro-geochemical flow was developed, which will serve as a basis for more detailed analyses in subsequent stages.

Key words: Brazilian Semiarid Region; Sedimentary Aquifer; Excess Air; Paleotemperature; Groundwater Flow.

A first look at the dissolved gas composition of offshore freshened groundwater (IODP 501, New England continental shelf, USA)

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Freshened porewater in marine sediments (characterized by its lower salinity relative to the overlying seawater) has been recognized since the 1960s and is referred to as offshore freshened groundwater (OFG) [1–2]. Evidence from scientific ocean drilling and hydrological modeling suggests that 10^5 – 10^6 km³ of OFG may be stored within continental shelf aquifers worldwide [3]. The possibility that OFG constitutes a substantial, yet largely untapped, source of freshwater has stimulated growing scientific interest over the past decade, with important implications for coastal and marine hydrogeology, paleoclimate and paleoenvironmental reconstructions, and groundwater-dependent coastal ecosystems. However, the origin(s), emplacement history, and dynamics of these shoreline-crossing groundwater systems remain poorly constrained.

Salinity data from the Atlantic continental shelf off New England have indicated that the present-day freshwater/saltwater interface is far out of equilibrium with modern sea-level conditions [4]. To shed light on the origin and evolution history of this hydrological system, the IODP Expedition 501 “New England Shelf Hydrogeology” cored and sampled the subseafloor offshore Massachusetts, USA (<https://www.ecord.org/expedition501/>), from May to August 2025. For the first time, they conducted a series of successful pumping tests on OFG bodies located up to 400 m below the seafloor. To monitor these pumping tests, we used a miniRuedi to measure dissolved gas concentrations in real time, enabling (i) assessment of the representativeness of samples collected by the multiple scientific teams, (ii) evaluation of the hydraulic confinement of the targeted aquifers, and (iii) qualitative estimation of groundwater residence times. The fun started right with the very first pumping test, when we realized that the best we were going to get was brackish water packed with fine particles.

This presentation will walk you through the motivation, strategy, and challenges of this project. We will share some of the general results and preliminary interpretations covering both the technical side of the pumping tests and the preliminary scientific insights they have revealed about the aquifers we explored.

[1] Kohout (1964), The flow of fresh water and salt water in the Biscayne aquifer of the Miami area, Florida Rep., USGS Water Supply, Washington, DC.

[2] Micallef, Person, Berndt, Bertoni, Cohen, et al. (2021), Offshore freshened groundwater in Continental margins, *Reviews of Geophysics*, 59(1), doi:10.1029/2020rg000706.

[3] Zamrsky, Essink, Sutanudjaja, Van Beek, and Bierkens (2021), Offshore fresh groundwater in coastal unconsolidated sediment systems as a potential fresh water source in the 21st century, *Environmental Research Letters*, doi:10.1088/1748-9326/ac4073.

[4] Person, M., Dugan, B., Swenson, J. B., Urbano, L., Stott, C., Taylor, J., & Willett, M. (2003). Pleistocene hydrogeology of the Atlantic continental shelf, New England. *Geological Society of America Bulletin*, 115(11), 1324-1343.

Beyond Stationary Models: A Practical Approach to Mixing Ratios and Travel Time in Alluvial Aquifers

Théo Blanc, Friederike Curre, Morgan Peel, Yama Tomonaga, Daniel Hunkeler, Oliver Schilling, Rolf Kipfer, Philip Brunner

University of Neuchâtel / Eawag / University of Basel

Understanding travel times and mixing dynamics in alluvial aquifers is essential for the effective management of riverbank filtration systems, especially under non-stationary conditions caused by restoration works or high-discharge events. Tracer studies are widely used to characterize such systems, but the resulting data are often interpreted using either overly simple stationary models that fail to capture dynamic behavior, or complex numerical simulations that are computationally intensive. In this study, we present a streamlined modeling framework using gas tracers to characterize surface water–groundwater (SW–GW) mixing and groundwater age in dynamic aquifer conditions. The framework integrates three components: (1) a binary mixing model, (2) an excess air model, and (3) a radon ingrowth model. Together, these tools allow us to resolve short-term variability in a highly dynamic river–aquifer system.

We demonstrate this approach using data collected from an alluvial plain in Bern, Switzerland, where drinking water is abstracted. During the monitoring period, restoration works were carried out in the nearby river. We combined continuous gas monitoring with miniRUEDI and Rad-7 at a groundwater well to track variations in mixing processes and apparent groundwater age over several months. This modular and low-complexity approach is transferable to other sites where understanding mixing and travel times is critical, such as drinking water abstraction zones or managed aquifer recharge systems.

Tracking CO₂ injection to an aquifer for CCS in Iceland

Matthias S. Brennwald, Jonas Simon Junker, Chuan Wang, Rolf Kipfer, Anne Obermann, Martin Voigt, and Alba Zappone

Eawag

Mineral sequestration of CO₂ in basalt aquifers is a promising pathway for permanent carbon storage. Within the DemoUpStorage R&D project, CO₂ was dissolved in saline groundwater and co-injected with helium as an inert tracer into a deep basalt aquifer at a pilot injection site near Helguvik, Iceland. We tested a combination of two novel geochemical and geophysical tools to monitor the subsurface CO₂ dynamics. Real-time, on-site monitoring of dissolved gases was conducted over one year using a gas-equilibrium membrane-inlet mass spectrometer (GE-MIMS). Electrical resistivity tomography (ERT) was used to track changes in aquifer resistivity before and after injection. GE-MIMS measurements revealed a clear increase in He concentrations downstream of the injection point, confirming fluid transport. However, no corresponding CO₂ increase was observed, indicating substantial CO₂ retardation relative to the fluid transport. No anomalies in He or CO₂ were detected in the overlying freshwater aquifer, indicating minimal upward migration. ERT data showed localized resistivity decreases suggesting basalt dissolution near the injection borehole. Together, GE-MIMS and ERT provided complementary, cost-effective insights into CO₂ transport, reaction, and storage processes in the basalt aquifer. These tools enhance monitoring and assessment capabilities, supporting the development of safe and effective geological CO₂ storage.

Estimating ebullition from wetlands using porewater gas concentrations

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Civil and Environmental Engineering, MIT, USA

Methane is transported from wetlands by a variety of processes: ebullition (exsolution of bubbles that rise to the surface), flushing (advection by pore water discharging into streams), and diffusion (both into plant roots and out through the surface). The competition between these processes leads to different porewater concentrations of dissolved gases. We hope to combine models of these competing processes with measured porewater gas concentrations in different wetlands to estimate methane fluxes from these wetlands. In particular, we hope to estimate ebullition of methane from measurements of noble gases in several wetlands in the eastern US. We also plan to use measured porewater gas concentrations to test hypotheses about why carbon dioxide concentrations are much lower in tropical peatlands than northern peatland.

Establishment of a new Groundwater Dating Laboratory at TU Dresden

Diana Burghardt, Paul Fremdling

TU Dresden, Institut für Grundwasserwirtschaft

Groundwater is the most important source of drinking and industrial water worldwide. Groundwater models and average residence times of groundwater are needed to assess usable groundwater reserves and their sustainable management. The only reliable method for dating sustainably usable groundwater is the analysis of helium isotopes dissolved in the water. Funding from the EFRE-InfraProNet 2021-2027 support programme and the Free State of Saxony will be used to establish a laboratory for groundwater dating using helium isotope analysis at the Institute for Groundwater Management at TU Dresden from 2026 to 2027. In addition to the funds for the system of all devices, personnel funds are also available. The objectives of the funding are (i) the technological establishment of this laboratory, (ii) the verification of helium isotope analysis of the new system in comparison to that of the helis laboratory at the University of Bremen, (iii) the validation of helium isotope analysis of the new system on representative groundwater samples in Saxony, and (iv) the development of a web map of sustainably usable groundwater reserves in Saxony.

Origin of fluid emissions along crustal to lithospheric faults in an extensional setting, Betic Cordillera, Spain

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Universit   de Pau et des pays de l'Adour

The Betic Cordillera (SE Spain) has experienced a complex geodynamic history involving crustal thinning related to slab retreat. This deep lithospheric dynamic was accompanied by alkaline to calc-alkaline volcanism and the exhumation of metamorphic domes. These processes feed an active fluid system, the respective contributions of which remain poorly constrained. However, understanding and quantifying these deep fluid circulations has major implications for geothermal exploration and natural gas migration. Several crustal to lithospheric-scale faults in the Internal Betics have accommodated most of the deformation. These active structures (Lorca earthquake, Mw 5.2) are associated with significant hydrothermalism, as indicated by water temperatures reaching up to 65   C, and may act as preferential pathways for fluid and heat migration. We sampled fluids (water and gas when bubbles were present) from 14 thermal springs with temperatures ranging from 15 to 65   C. Gas species were measured in the field using a portable mass spectrometer (MiniRUEDI) along these strike-slip faults. The C  diz-Alicante Fault releases fluids rich in N₂ (86–96%), with CO₂ contents between 1 and 5%, and O₂ between 0.5 and 10%. Along the Alpujarras Fault, CO₂ dominates (63–91%) with lower amounts of N₂ (5–28%). The lithospheric Carboneras Fault emits fluids rich in N₂ (91%), with low proportions of O₂ (3%) and CO₂ (3%). Alhama de Murcia, the most active fault of the system, mainly releases CO₂ (52%) and N₂ (43%). This result helps us in discussing fluid origin in this extensive context.

Vermifiltration gas emissions and mass balances

Kayla Coppens

Eawag / University of Geneva

The miniRUEDI was used with a static flux chamber to quantify CO₂ and CH₄ emissions from 6 laboratory-scale vermifilters treating blackwater and brownwater at varying hydraulic loading rates. The gas emissions were further used to explore the impact of wastewater type and hydraulic loading on nitrogen and carbon mass balances.

Continuous miniRUEDI measurements reveal gas-seismic interactions at Lavey-les-Bains

Sébastien Giroud, Matthias Brennwald, Rolf Kipfer

Eawag

The link between dissolved gas concentrations in geological fluids and seismic activity remains debated. Reported correlations between gas composition changes and earthquakes are typically anecdotal and event-specific, while systematic, long-term datasets are rare [1]. To overcome this limitation, we deployed a portable gas equilibrium membrane-inlet mass spectrometer (miniRUEDI; [2]) at the geothermal springs of Lavey-les-Bains (50 °C to 65 °C), located in a seismically active region of Switzerland. The instrument continuously recorded dissolved gas concentrations (He, Ar, Kr, N₂, O₂, H₂, CH₄, and CO₂) over more than three years, providing one of the longest high-frequency datasets available. We use these time series for a critical assessment of the temporal correlation between seismic events and gas evolution in geothermal fluids.

[1] Toutain and Baubron (1999), *Tectonophysics*, 304, 1–27

[2] Brennwald et al. (2016), *ES&T*, 50, 13455–13463

Zimbabwe-Headwater Catchment Response to Extreme Rainfall Activities-Meeting on sampling of hydrological extreme events

Webster Gumindoga

University of Zimbabwe

Extreme rainfall events are becoming increasingly frequent across Zimbabwe, yet their effects on headwater catchment hydrology remain poorly quantified. Understanding rainfall–runoff–groundwater interactions under such events is essential for water resources planning, particularly in convective rainfall-dominated systems. This study assessed the hydrological and isotopic response of three representative headwater catchments—Domboshava (Mazowe), Marondera (Manyame), and the University of Zimbabwe (Gwebi)—during the 2023/2024 wet season. Stable isotopes of oxygen ($\delta^{18}\text{O}$) and hydrogen ($\delta^2\text{H}$) were analysed in rainfall, surface water, and groundwater, while Radon-222 (^{222}Rn) was used as a tracer of subsurface flow dynamics. Sampling was conducted at event and intra-event scales, with rainfall collected up to three times daily and groundwater monitored through logging wells. Early-season rainfall at the University of Zimbabwe site exhibited isotopic enrichment ($\delta^{18}\text{O} = 3.06\text{‰}$; $\delta^2\text{H} = 17.48\text{‰}$) and low d-excess (-7‰), indicating subcloud evaporation and limited recharge potential. Intra-event sampling revealed “ Λ -shaped” isotopic profiles, reflecting the temporal variability within convective storm systems. Distinct isotopic signatures were observed among rainfall, surface water, and groundwater, underscoring episodic recharge patterns. Variability in ^{222}Rn concentrations indicated contrasting subsurface flow paths and groundwater residence times across sites. The study highlights the heterogeneity of recharge processes in Zimbabwean headwater catchments and the influence of convective rainfall on water partitioning. Data limitations due to manual sampling and equipment gaps remain key challenges. Future work will focus on improved monitoring networks, recharge zone mapping, and long-term isotope-hydrology integration for water resource resilience under extreme rainfall conditions. Keywords: Stable isotopes, Convective rainfall, Groundwater recharge, Headwater catchment, Radon-222, Zimbabwe.

Noble gases – suitable tracers to optimize the productivity of a managed aquifer system?

Jakob Helbing, Marie-Sophie Maier

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The Hardhof groundwater field is essential for a secure drinking water supply in the city of Zurich. The city's urban infrastructure surrounds the groundwater field and thus limits its local extension. To increase the natural groundwater pumping capacity and guarantee safe drinking water supply at any time, the groundwater field is artificially managed. Riverbank-filtered Limmat water is pumped across the field and injected into the ground along the most vulnerable side towards heavy traffic infrastructure. To determine the necessary amount of water injected into the ground, a sophisticated real-time hydrological numerical model is employed. For an optimal performance, this model needs to be calibrated and evaluated. One possibility to do so is identifying real flow paths with tracer experiments and comparing these results with the output of the numerical model. In a groundwater protection zone however, nonhazardous tracers are vital. The groundwater field Hardhof offers a possibility to test the assumption, whether noble gases can be used as tracers in large scale managed aquifer recharge (MAR) systems. Approximately 100 boreholes within the groundwater field can be used both as injection points and measurement locations for noble-gas tracers. Therefore, different setups of distance and spread can be well tested. In our presentation, we will discuss experimental design and first estimations of this project.

TBD

Rolf Kipfer

AFFILIATIONS

TBD

Installation of a noble gas laboratory in Pau

Benjamin Lefeuvre, Anne Battani

LFCR - UPPA

oble gases (He, Ne, Ar, Kr, Xe) are excellent natural tracers for studying geological processes, as their fully filled outer electron shells make them chemically inert. A new noble gas laboratory has been established in Pau through joint funding from TotalEnergies and the University of Pau, dedicated to the characterization of these gases in natural samples. Two mass spectrometers (an Argus VI and a Helix SFT from Thermo Fisher Scientific) have been acquired and installed. Prior to sample introduction into these instruments, a custom purification and separation line was developed. This line enables the removal of reactive gases through a series of physico-chemical traps, such as cold fingers filled with activated charcoal, Zr-Al getters, and titanium sublimation pumps (TSPs). Once purified, the remaining noble gases are cryogenically separated and introduced into the mass spectrometers for isotopic analysis.

In addition, a quadrupole mass spectrometer (QMG 250, Pfeiffer Vacuum) has been installed to monitor the purification process and to analyze major gaseous components. This instrument is similar to the MiniRuedi mass spectrometer and is equipped with two detectors: a Faraday cup and an electron multiplier allowing the quantification of gases at both high and low concentrations.

Dissolved Gas Monitoring at the Bedretto Underground Laboratory: Responses to Seismic Stimulation

**Alexandra Lightfoot, Matthias Brennwald, Valentin Gischig, Alba Zappone,
Rolf Kipfer, Stefan Wiemer**

SED ETH Zürich

Gas Monitoring at the Bedretto Underground Laboratory: Responses to Seismic Stimulation In hydraulic stimulation experiments, such as those used in seismic and geothermal energy studies, monitoring dissolved reactive and noble gases in groundwater can provide key insights into fluid transport and redistribution. While tracers added to injection water reveal specific flow paths, dissolved gas monitoring in general captures a broader picture of how fluids are mobilised or redirected under pressure. At the Bedretto Underground Laboratory for Geosciences and Geoenergy (BULGG), we tested the geochemical response during an experiment designed to trigger a controlled magnitude-zero earthquake. High-pressure water (up to 20 MPa) was injected into borehole ST1, with krypton added as a tracer. A nearby borehole (ST2), 44 m away and discharging continuously, was equipped with a miniRuedi to measure dissolved He, Ar, Kr, N₂, O₂, and CO₂ in real time, complemented by parallel radon monitoring. This setup enabled us to track both immediate gas responses to pressure and delayed tracer migration. Dissolved gas signals at ST2 revealed a clear response once injection began. N₂ and Ar increased in terms of partial pressure, while He decreased, showing that injection resulted in mobilised or redirected younger, less radiogenic fluids toward ST2 well before the Kr tracer arrived. Radon activity increased only after tracer breakthrough, consistent with fracture opening and fluid movement under stress. These findings highlight dissolved gas monitoring as a sensitive tool for probing the interplay between stress, fractures, and fluids, particularly in the context of hydraulic stimulation.

New tool for in-situ gas measurements in low flow porous media

C. Marion, C. Ebi, S. Bloem, M.S. Brennwald, R. Kipfer

Eawag

Gases, including noble gases, are powerful tracers in environmental science, widely used to study groundwater flow dynamics and surface–subsurface water interactions ([1], [2], [3], [4], [5], [6]). Available technologies ([7], [8]) allow such measurement set-up only for high water flow environmental systems (L/min to L/s); systems in which the extraction of liters of water per minute would not disturb the systems equilibrium. In the meanwhile, in-situ gas measurements in low-flow porous media (L/h), such as soil unsaturated and saturated zones or plant tissues, are technically challenging, even though these systems play a critical role for water and CO₂ exchange on the global scale. We present a robust, field-deployable gas probe for long-term, in-situ gas monitoring in low-permeability or low-flow environments. Originally developed for in-vivo gas monitoring in tree xylem ([9]), this method enables non-destructive, high-resolution measurements applicable to a variety of porous media, including soil, sediment, rock, and biological tissue.

- [1] Brennwald, M. S. et al. (2022). *Frontiers in Water*, 4, 107.
- [2] Blanc, T. et al. (2024). *Water Research*, 254, 121375.
- [3] Schilling, O. S. et al. (2017). *Water Resources Research*, 53(12), 10465–10490.
- [4] Schilling, O. S. et al. (2021). *Water Resources Research*, 57(2).
- [5] Peel, M. et al. (2022). *Chemical Geology*, 599, 120829.
- [6] Giroud, S. et al. (2023). *Frontiers in Water*, 4, 1032094.
- [7] Brennwald, M. S., et al. (2016). *Environmental Science and Technology*, 50.
- [8] Maechler, L. et al. (2014). *Chimia*, 68(3), 155–159.
- [9] Marion, C. et al. (2024). *Tree Physiology*.

Noble gases as tracers for groundwater in a big high-altitude lake

Paul Moser Röggl, The ICDP Nam Core Team, Torsten Haberzettl, Hendrik Vogel, Rolf Kipfer

Eawag

Nam Co (30°42'N, 90°33'E; 4712 m a.s.l), consisting of a deeper main basin (max depth: 100m) and a shallower eastern basin (max depth: 50m), is one of the highest saline lakes in the world and is thought to exhibit substantial groundwater–lake interactions (see e.g. Ren et al., 2025). Recent mass balance estimates suggest that up to 25% of the lake's annual water input originates from lacustrine groundwater discharge (Ren et al., 2025). Terrestrial groundwater typically carries an excess air signature, characterized by oversaturation of atmospheric noble gases, while dissolved gases in surface water bodies are generally expected to remain in atmospheric equilibrium (Brennwald et al., 2013). However, noble gas measurements from the main basin of Nam Co reveal a distinct excess air signal at 50 m depth, with ΔNe oversaturations of up to 30%. In addition, water at this depth shows a measurable tritium age, which is unexpected for a dimictic lake with seasonal overturn. We interpret these observations as direct measurable evidence for permafrost-derived groundwater discharge into the eastern basin, followed by lateral transport toward the central basin via horizontal, density-driven currents.

Brennwald, M.S. et al., In: Burnard, P. (Ed.), *The Noble Gases as Geochemical Tracers*. Springer, Berlin, Heidelberg, pp. 123–153.

Ren, Wenhao et al., *Journal of Hydrology*, 662, Part B, 2025, 133956

Field Deployment of a miniRUEI for Long-term Gas Monitoring in the Mt. Fuji Volcanic Aquifer System

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A portable quadrupole mass spectrometer (miniRUEI) was installed at a drinking water well intersecting the Fuji-Kawa-Kako fault zone — one of Japan's most active fault systems — within the Mt. Fuji catchment. The system continuously monitors noble and reactive gases in both ambient air and groundwater using the gas-equilibrium membrane inlet mass spectrometry (GE-MIMS) approach. This long-term deployment, ongoing since 2023, provides new insights into hydrological and geochemical dynamics within Mt. Fuji's volcanic aquifer system. Distinct variations in CO₂, O₂, and ⁴He concentrations were observed following major precipitation events, indicating the infiltration of newly recharged meteoric water. Additional transient gas anomalies coincided with regional seismic activity, suggesting a potential link between crustal stress changes and subsurface gas transport. Despite maintenance requirements due to water condensation and membrane module clogging, the system demonstrated stable, long-term operation. Overall, the results highlight the potential of continuous gas monitoring for resolving coupled hydrological, geochemical, and tectonic processes in active volcanic environments.

From residence time to flow rates: insights from $^{234}\text{U}/^{238}\text{U}$ activity ratios and REE patterns of the ancient Kurnub Group aquifer

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Measurements of uranium isotopes and rare earth element (REE) compositions were used to evaluate water-rock interaction processes, residence times, mixing, and flow rates in the Kurnub Group Aquifer (KGA), a deep confined regional sandstone aquifer extending from the Sinai Peninsula (Egypt) to the Negev Desert (Israel). The study involved developing a methodology that allowed to separate both U and REE from the brackish groundwater using a single column. Groundwater was sampled along two main flow paths of the KGA: East (n=9) and North (n=8), and from the overlaying Judea Group Aquifer (JGA; n=2). Overall, samples along the KGA flow paths show consistent low REE concentrations (ppt levels) and patterns, with low LREE and HREE and high MREE. The KGA samples have high $^{234}\text{U}/^{238}\text{U}$ activity ratios (AR of up to 8.37), with a systematic exponential decrease in AR downflow as a function of distance and head elevation. This pattern suggests that AR decreases downflow mainly by radioactive decay, enabling the estimation of its long-term average flow rate to 24 cm/yr, similar to recently obtained ^{81}Kr -based estimations. An exceptionally high AR is observed near a major fault zone, suggesting that the highly enriched ^{234}U may originate from this deep-seated active deformation zone, whereby fluid interactions with fractured rocks lead to the preferential leaching of the ^{234}U isotope. Assuming that the computed average flow rate is representative also for the upper parts of the aquifer in Sinai, the uranium isotope-based maximum residence time of groundwater in the KGA is estimated to be 1.3 Myr.

Tracing Hidden Connections: Multi-Tracer Evidence of Managed and Natural Controls on Groundwater–Surface Water Contributions to the Eerste River Estuary, South Africa

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Estuarine wetlands in Mediterranean Climates are highly sensitive to variations in water availability and quality, which influence their ecological services. The Eerste River Estuary in the Western Cape, South Africa, is vulnerable to the combined pressures of recharge variability, urbanisation, and water transfer schemes. In this data-limited, urbanised catchment, a multi-tracer approach was applied to investigate the spatio-temporal contributions of water sources (rain, surface water, and shallow and deep groundwater) to the estuary. Major ions, stable isotopes (ffⁱ¹⁸O, ffⁱ²H), tritium, and dissolved gases (N₂, O₂, CH₄, CO, He, Ne, Ar, Kr, Xe) were analysed from 17 rainwater, 83 surface water, and 33 groundwater (shallow and deep aquifers) sites between 2022 and 2024 dry and wet seasons. Hydrochemical facies evolve from the freshwater Na–Cl type in the headwater recharge to Na–Cl–Ca–HCO₃ type near the estuary, reflecting a spatial gradient controlled by precipitation, evaporation and anthropogenic influences, with greatest fluctuations in the estuary chemistry. Isotopic data indicates evaporation losses of the estuary are buffered by a dual-water system: short-term variability is dominated by upstream surface water transfer schemes, while longer-term resilience depends on groundwater of contrasting ages and origins. Groundwater tritium levels are uniformly low (<0.5 TU), indicating predominantly pre-modern recharge and long residence times, though some sites (0.5–1.7 TU) suggest minor mixing with modern water. The presence of crustal-derived helium (³He/⁴He R/R_a = 0 – 0.04) and ⁴He-excess air further supports the dominance of old groundwater with limited modern recharge. These findings show that although the estuary is buffered by managed inflows, it remains vulnerable to climatic variability and abstraction pressure. The multi-tracer framework has refined the conceptual understanding of complex hydrodynamic systems and guides for sustainable monitoring and management of estuarine wetlands.

Seasonality of Excess Air Signatures Under Varying Flow Conditions in a Sub-Alpine Fractured Bedrock Aquifer

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We report on a tracer study to constrain groundwater dynamics in an alpine groundwater system near Crested Butte (Colorado, USA), dominated by snowmelt patterns. The hydrograph in this area, underlain by fractured shale, shows two flow periods: a base-flow period (BP) lasting from late summer to early spring, where no recharge occurs and the water level is 3-4 m below land surface; and a recharge period (RP) starting in late spring caused by snowmelt, leading to a rapid rise in water level by 1.5 m, which then declines until late summer. Dissolved noble gases were sampled at different times from a 10 m deep bedrock well at the bottom of a hillslope transect close to the East River. We observe different DeltaNe signatures during both flow periods: positive DeltaNe during RP and negative DeltaNe during BP. Our preliminary results suggest a link to (1) the hydrostatic pressure and (2) prevalent CO₂, N₂, and CH₄ subsurface production caused by bedrock weathering. Lower hydrostatic pressure during BP results in exsolution of oversaturated gases, which may be enhanced by continuous subsurface gas production. At higher hydrostatic pressure during RP, the positive DeltaNe from recharge may prevail. Our results highlight the complexity and seasonality of groundwater dynamics in snowmelt driven mountainous fractured bedrock aquifers, furthered by additional processes like subsurface gas production.

Noble gases as artificial tracers for routine hydrogeological investigations

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Eawag

Noble gases are increasingly adopted as artificial tracers for hydro(geo)logical studies, thanks to advances in portable, high-resolution dissolved gas measurement technologies [1-3]. Their inert, non-toxic, and invisible nature makes them ideal for sensitive environments, such as rivers and drinking water catchment areas, where maintaining water quality and appearance is essential. The application of gases in aqueous environments poses unique challenges compared to traditional tracers, as potential degassing needs to be accounted for, and ideally avoided, during tracer injection. We present a straightforward method for preparing and injecting tracer solutions with accurately controlled dissolved gas concentrations, whilst avoiding tracer loss. A two-step degassing-pure gas saturation approach enables rapid preparation of high-concentration tracer solutions using readily available materials. We applied this method in a riverbank filtration setting to perform well-to-well tracer tests using helium-4 (He-4) and krypton-84 (Kr-84). Known quantities of both tracers were injected into observation wells upstream of a pumping well. A portable mass spectrometer ("miniRUEDI" GE-MIMS [3]) was used for continuous monitoring in the pumping well. Breakthrough curves of He-4 and Kr-84 enabled precise determination of groundwater flow velocities, travel times, and tracer recovery. These findings illustrate how noble gases complement traditional tracer methods in hydrogeological investigations. The method is adaptable to other gases (e.g., Ne, Kr, Xe, light hydrocarbons), significantly expanding the range of tracers available in groundwater management contexts. 1 : Brennwald et al., 2022, *Front. Water* (4) 2 : Blanc et al., 2024, *Wat. Res.* (254) 3 : Brennwald et al., 2016, *Env. Sci. Techn.* (50)

Investigating gasgeochemical production during the preparatory phase of dynamic brittle failure in Rotondo granite in the laboratory

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Laboratory geochemistry studies enhance our understanding of earthquake precursors. Controlled experimental conditions allow researchers to observe how microcracks, fluid flow, and mineral reactions influence geochemical signals, including gas emissions and isotope distributions. These experiments can identify specific geochemical changes, such as the release of radon or noble gases, associated with rock fracturing. Such insights are crucial for developing indicators of large events while calibrating geochemical signals, establishing links between these signals, stress states, and damage levels within rocks.

This study investigates the use of helium, argon, and CO₂ as real-time geochemical tracers for brittle deformation in crystalline rocks, focusing on Rotondo granite from the Bedretto Laboratory. These gases, stored in mineral lattices or fluid inclusions, are released during microstructural damage; however, their behavior during progressive rock failure remains poorly understood. Uniaxial compression experiments monitored gas variability with a miniRuedi quadrupole mass spectrometer, complemented by spatial strain measurements using Distributed Fibre Optic Strain (DFOS) sensing, acoustic emission (AE) monitoring, and post-mortem X-ray computed tomography (XRCT) of fracture networks. The experimental design aims to identify distinct deformation stages, from crack initiation to macroscopic failure, and link these to transient noble gas signals. Preliminary results reveal stage-specific fluctuations in helium and CO₂, supporting the hypothesis that gas emissions reflect structural damage evolution. This research advances the integration of mechanical and geochemical monitoring in rock mechanics, with important implications for underground stability assessment and early warning systems in geotechnical and seismic environments.

Year-Long miniRUEDI Gas Time Series Reveal Shutdown-Driven Excess Air and Water-Quality Impacts

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Eawag

Managed aquifer recharge (MAR) operations can unintentionally modulate subsurface gas dynamics, with consequences for redox conditions and groundwater quality. In this study, we deploy a portable mass spectrometer (miniRUEDI) in an observation well at the Hardwald MAR scheme near Basel, Switzerland, and acquire continuous noble- and reactive-gas records (He, Ar, Kr, N₂, O₂, CO₂) over a one-year period. The instrument enabled unattended, on-site measurements with percent-level precision and high temporal resolution, well-suited to capture transient events. During the study, MAR operations ceased eight times for periods ranging from hours to several days. Each cease produced a potential characteristic multiphase gas response: (i) rapid shifts in dissolved gas ratios consistent with bubble entrapment and dissolution upon re-wetting, i.e., formation of “excess air”; (ii) step-changes in O₂ and N₂ that propagated on operational time scales; and (iii) knock-on effects on redox-sensitive species (CO₂, CH₄) indicative of short-term biogeochemical reorganization. By combining noble-gas diagnostics of excess air with reactive-gas trends, we separate physical recharge effects from biogeochemical responses and quantify how shutdowns alter effective recharge signatures and residence-time proxies. These findings highlight portable mass spectrometry as a process-control tool for MAR, linking operational decisions to subsurface gas dynamics and water-quality outcomes in near-real time. Our results generalize established concepts of excess-air formation in porous media to the operational rhythms of MAR and provide actionable guidance for minimizing adverse water-quality swings during start/stop cycles.

Continuous in situ-based gas measurements for volcanic gas monitoring, developing a membrane degasser, and profiling gases in a stratified pond

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I will share a brief and informal overview of different applications of in-situ miniRUEDI measurements by our group over recent years and present a few avenues for potential future developments and collaboration. These include (i) pilot results for monitoring of CO₂-dominated volcanic gas emissions in bubbling springs, (ii) testing of a prototype membrane-based degasser, and (iii) depth profiling of CH₄ and other dissolved gases in a highly stratified brackish pond.

For part (i): Previous work by our group and others has shown that noble gases in surface CO₂ degassing sites (e.g., hot and cold springs) show high variability both spatially (e.g., two bubbling plumes within a meter might vary widely in noble gas composition) and temporally (e.g., repeat sampling of the same site has revealed considerable variation over time). In these systems, where Ne, Ar, Kr, and Xe are largely dominated by a groundwater (i.e., atmospheric) component that has degassed into an upwelling CO₂ gas phase, noble gas elemental ratios typically vary between air-like and air-saturated-water-like values, while recent heavy noble gas isotope measurements have shown extensive kinetic fractionation associated with intermediate elemental ratios. Continuous monitoring of noble gases in such systems may shed light on the mechanisms of water-gas interaction, which in turn might inform larger-scale relationships between volcanic and hydrogeologic systems in areas prone to eruption.

For part (ii): Together with a group of undergraduate student fellows, we developed a simple single-membrane degasser for the collection of sufficiently large quantities of gas to analyze the isotopic composition of rare gas species (e.g., Hg) in water. To evaluate the performance of a single-membrane degasser with respect to individual gas species, with the goal of upscaling such a system to achieve quantitative degassing for higher solubility rare gases, we carried out tests in WHOI's AVAST center in a saltwater tank in which a miniRUEDI GE-membrane was placed downstream of the degasser to constrain gas-specific extraction efficiencies to tune a model that could be extended to other gases.

For part (iii): To investigate physical and biogeochemical dynamics of a brackish, meromictic pond in Falmouth, MA, we carried out continuous dissolved gas measurements using a miniRUEDI aboard a small dinghy. Using a submersible pump that we gradually lowered to different depths, we analyzed noble gases, N₂, O₂, and CH₄, producing a data set that may be useful in ultimately constraining the flux of CH₄ through the pond, while also providing an interactive educational opportunity in both mass spectrometry and environmental biogeochemistry to a group of undergraduate students.

A (very) portable Gas Equilibrium Membrane Sampling System (GEMSS)

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The miniRUEDI has proved to be a valuable tool to monitor continuously and in real time the gas dynamics in environmental systems. While mid- and long-term observations imply generally fixed installations with a relatively constrained effort in terms of logistics, the spatial screening of the gas composition in surface waters and groundwater requires the transport of miniRUEDI with the respective GE-MIMS (and a power source, e.g. a generator) to the different sampling locations. This can be sometimes problematic, as the instrument weight of about 30 kg (not including pumps, generator ...) makes it "portable" only to a certain extent. To address this limitation, and to simultaneously improve on the long standing limitation of samples taken in copper tubes not being pre-screenable on the miniRUEDI prior to time-consuming noble gas isotope analyses, we developed a small portable Gas Equilibrium Sampling System (GEMSS aka "James") that can be taken to the field easily and allows for a sufficient amount of gas to be sampled for multiple analyses on a single sample. The system uses the same concept of gas equilibrium of the conventional GE-MIMS. However, instead of functioning as an inlet for miniRUEDI measurements, James allows acquisition of gas samples for subsequent analyses in reusable containers. James is relatively light and fits in a backpack, it not only allows replicate measurements on a miniRUEDI, but the respective samples can further be used also for other types of analyses (e.g., conventional noble gas isotope analyses on magnetic sector mass spectrometers).